

MTH203: Sequences and series

Lecture 11

1. REMARKS ON THE BOLZANO-WEIERSTRASS THEOREM

In the previous lecture, we proved that any bounded sequence in $\mathbb{R}$ has a convergent subsequence. This leads us to define the following:

Definition 1. We say that a metric space has the *Bolzano-Weierstrass property* if every bounded sequence is convergent.

See Lecture 8 for the definition of bounded sequences in a metric space.

We have proved that $\mathbb{R}$, with its standard metric, has the Bolzano-Weierstrass property. Let us look at another example:

Example 2. Let k be any positive integer. Then, we define the *Euclidean metric* on $\mathbb{R}^k$ as follows:

Given $\mathbf{x} = (x_1, \dots, x_k)$ and $\mathbf{y} = (y_1, \dots, y_k)$ in $\mathbb{R}^k$, we define the distance $d(\mathbf{x}, \mathbf{y})$ by

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \dots + (x_k - y_k)^2}.$$

When $k = 2$ or 3 , this is the usual distance that we are familiar with. We will not verify here that this is a metric space. (See the reading suggestions below.)

If we assume that this indeed a metric space, it is very easy to see that it has the Bolzano-Weierstrass property. We will not prove this explicitly, but we will sketch two proofs. (You may fill in the gaps to write a rigorous argument as an exercise.)

For our first sketch, let us assume that $k = 2$ for the sake of simplicity. Let $(\mathbf{x}_n)_{n \in \mathbb{N}}$ be a bounded sequence in $\mathbb{R}^2$. Then, one can show that the entire sequence can be contained in a closed rectangle. A *closed rectangle* in $\mathbb{R}^2$ is defined as a subset of the form $I \times J \subset \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$, where I and J are closed intervals in $\mathbb{R}$. Now, one can imitate the proof we have given for the Bolzano-Weierstrass theorem, with the rectangle playing the role of an interval. We divide the rectangle into 4 smaller rectangles, such that each of them has sides that are half the lengths of the sides of the original rectangle. At least one of these four rectangles will contain infinitely many terms of the sequence. Just as we had proved that the intersection of a nested sequence of closed intervals is non-empty, one can also prove that the intersection of a nested sequence of closed rectangles is non-empty. It is easy to complete the proof along these lines.

Now let us look at another proof. Again, let us take $k = 2$. Suppose we are given the bounded sequence $(\mathbf{x}_n)_{n \in \mathbb{N}}$. For each n , suppose $\mathbf{x}_n = (x_{1,n}, x_{2,n})$. Then we consider the sequence $(x_{1,n})_{n \in \mathbb{N}}$ in $\mathbb{R}$. First, one proves that this sequence is bounded. (We will leave this as an exercise.) Now, by the Bolzano-Weierstrass theorem for $\mathbb{R}$, we know that there exists a subsequence $(x_{1,n_k})_{k \in \mathbb{N}}$ which converges. Now, consider the sequence $(x_{2,n_k})_{k \in \mathbb{N}}$. This sequence is also bounded and one can choose a subsequence $(x_{2,n_{k_j}})_{j \in \mathbb{N}}$ which is convergent. Then, it is easy to check that the sequence $(\mathbf{x}_{n_{k_j}})_{j \in \mathbb{N}}$ is convergent.

Now, let us look at an example of a metric space that does not have the Bolzano-Weierstrass property.

Example 3. Let us define a distance function on $\mathbb{N}$ by setting $d(i, j) = 1$ if $i \neq j$. Then the sequence $(n)_{n \in \mathbb{N}}$ is a bounded sequence in $\mathbb{N}$ (with respect to this metric). However, it is clear that it does not converge.

2. SUMMATION OF SERIES

Definition 4. An *infinite series* (in $\mathbb{R}$) is a formal expression of the form $\sum_{i=1}^{\infty} b_i$, also written as $b_1 + b_2 + b_3 + \dots$, where $(b_i)_{i \in \mathbb{N}}$ is a sequence in $\mathbb{R}$.

Thus, loosely speaking, it is an infinite sequence of real numbers that we would like to add together. Of course, the addition of infinitely many real numbers has, a priori, *no meaning* at all. We have to assign some meaning to this expression. This is done as follows:

Definition 5. Let $\sum_{i=1}^{\infty} b_i$ be an infinite series in $\mathbb{R}$. For any positive integer n , the n -th *partial sum* of this series is defined to be $s_n := \sum_{i=1}^n b_i$. We say that the series *converges* if the sequence $(s_n)_{n \in \mathbb{N}}$ converges. If this is so, the expression $\sum_{i=1}^{\infty} b_i$ is used to denote the real number $\lim_{n \rightarrow \infty} s_n$. If the series does not converge, we say that it *diverges*.

Example 6. The series $\sum_{i=1}^n 1$ diverges. Its partial sums are $s_n = n$.

Example 7. Let x be any positive real number. Let $m.d_1d_2\dots$ denote its decimal expansion. Thus, m is a non-negative integer and for each i , d_i is an integer such that $0 \leq d_i \leq 9$. For any positive integer k , define $x_k = m + \sum_{i=1}^k d_i/10^i$. Then, clearly $(x_k)_{k \in \mathbb{N}}$ is a non-decreasing sequence. We saw in Lecture 6 that this sequence is bounded and its supremum is precisely equal to x . We also know that for a non-decreasing, bounded sequence, its supremum is also the same as its limit (see Lecture 8, Theorem 7). Thus, we see that the series $\sum_{i=1}^{\infty} d_i/10^i$ is convergent and $m + \sum_{i=1}^{\infty} d_i/10^i = x$.

Example 8. We now consider the series $\sum_{n=1}^{\infty} 1/n$. We will show that this series diverges. Let $s_n := \sum_{i=1}^n 1/i$. We will show that the sequence $(s_n)_{n \in \mathbb{N}}$ diverges.

The idea is to create a sequence $(a_n)_{n \in \mathbb{N}}$ such that $0 < a_i \leq 1/n$ such that the series $\sum_{n=1}^{\infty} a_n$ is easily seen to diverge. Then, if $t_n := \sum_{i=1}^n a_i$ are the partial sums of this series, the sequence $(t_n)_{n=1}^{\infty}$ will be seen to be an unbounded, strictly increasing sequence. Thus, it will follow that $(s_n)_{n \in \mathbb{N}}$ is also an unbounded sequence, and hence cannot converge.

We now construct the sequence $(a_n)_{n \in \mathbb{N}}$. For any positive integer n , there exists a unique non-negative integer k such that $2^{k-1} < n \leq 2^k$. Then, we set $a_n = 1/2^k$. By construction, $0 \leq a_n \leq 1/n$ for any n . Thus, it follows that the sequence $(t_n)_{n=1}^{\infty}$ of partial sums of the series $\sum_{i=1}^{\infty} a_i$ is a strictly increasing sequence. Also, we have $t_n \leq s_n$ for all n .

The initial terms of the sequence $(a_n)_{n \in \mathbb{N}}$ are as follows:

$$1, 1/2, 1/4, 1/4, 1/8, 1/8, 1/8, 1/8, 1/16, 1/16, \dots$$

Thus, we see that for any non-negative integer k , all the terms from a_{2^k+1} to $a_{2^{k+1}}$ are equal to $1/2^{k+1}$. There are precisely 2^k such terms. Thus, we have

$$a_{2^k+1} + a_{2^k+2} + \dots + a_{2^{k+1}} = 2^k/2^{k+1} = 1/2.$$

However, the left hand side is easily seen to be equal to $t_{2^{k+1}} - t_{2^k}$. Thus, we see that $t_{2^{k+1}} - t_{2^k} = 1/2$ for every non-negative integer k .

We see that $t_1 = 1$, $t_2 = 1 + 1/2$, $t_4 = 1 + 1/2 + (1/4 + 1/4)$, and so on. In general, if $k \geq 1$, Thus, for any positive integer $k \geq 1$, we have

$$\begin{aligned} t_{2^k} &= (t_{2^k} - t_{2^{k-1}}) + (t_{2^{k-1}} - t_{2^{k-2}}) + \dots + (t_2 - t_1) + t_1 \\ &= 1/2 + 1/2 + \dots 1/2 + 1 \\ &= k/2 + 1 \end{aligned}$$

Since $s_{2^k} \geq t_{2^k}$, we see that the sequence $(s_n)_{n=1}^{\infty}$ is unbounded, and so the series $\sum_{n=1}^{\infty} 1/n$ diverges.

3. READING SUGGESTIONS

I do not have a precise reference for the discussion in Section 1 of this lecture. For a discussion of the Euclidean metric on $\mathbb{R}^n$, see [2, Theorem 1.37]. The material related to infinite series is from Section 2.4 of the textbook ([1]).

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [2](#)
- [2] Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill (3rd edition), 1976. [2](#)